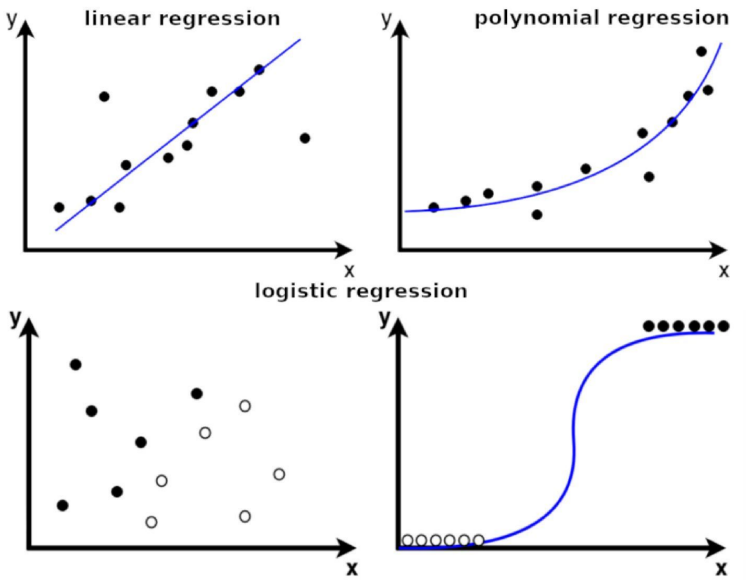
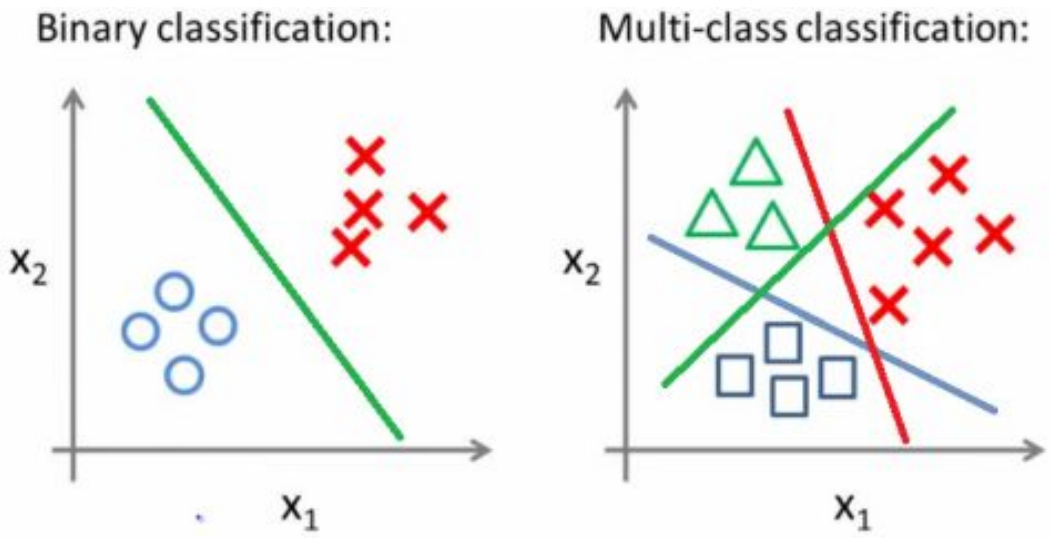
More About Classification Models

DATA 201 -- WEEK 8B
SPRING 2026

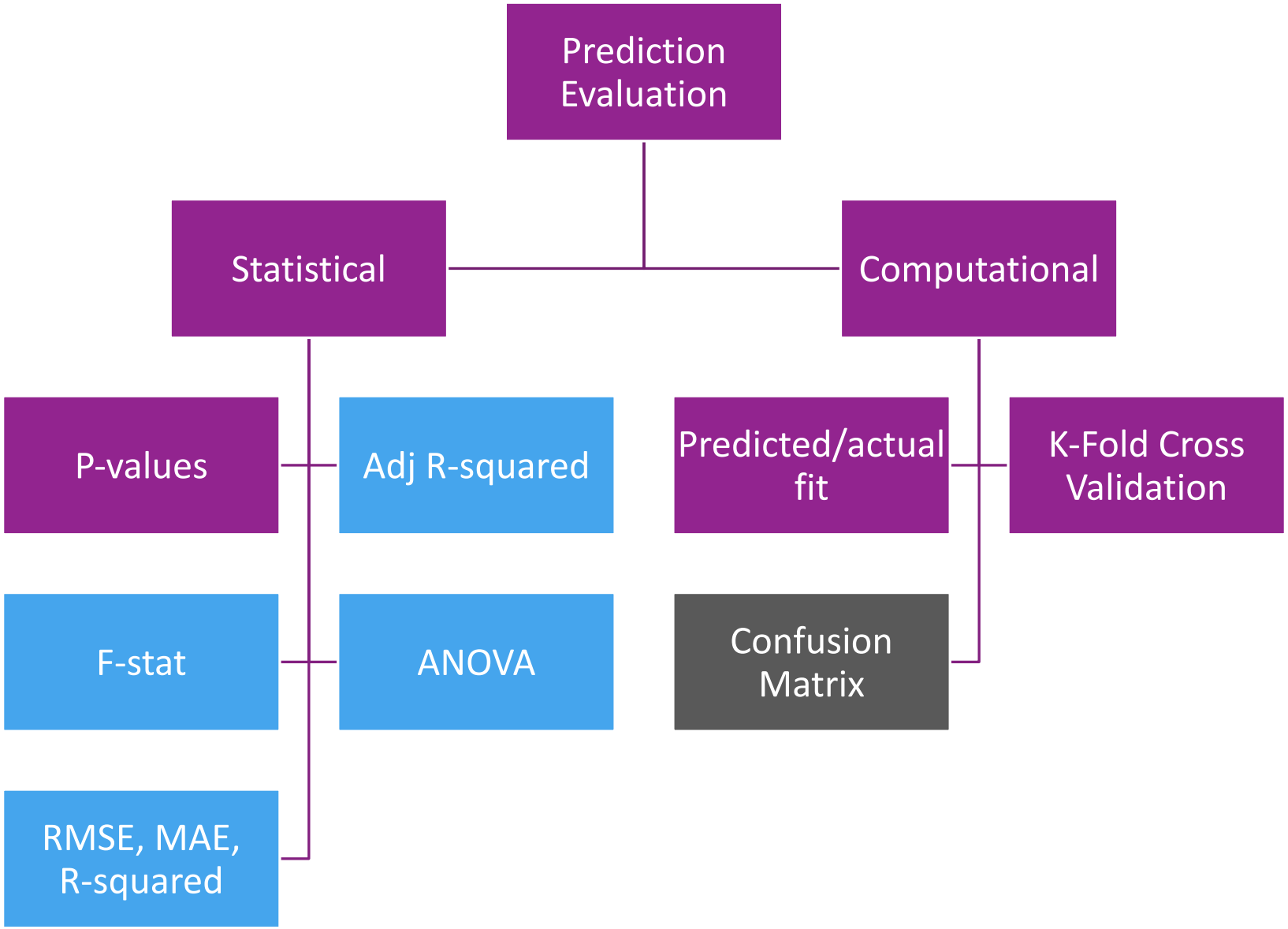
Topics Today

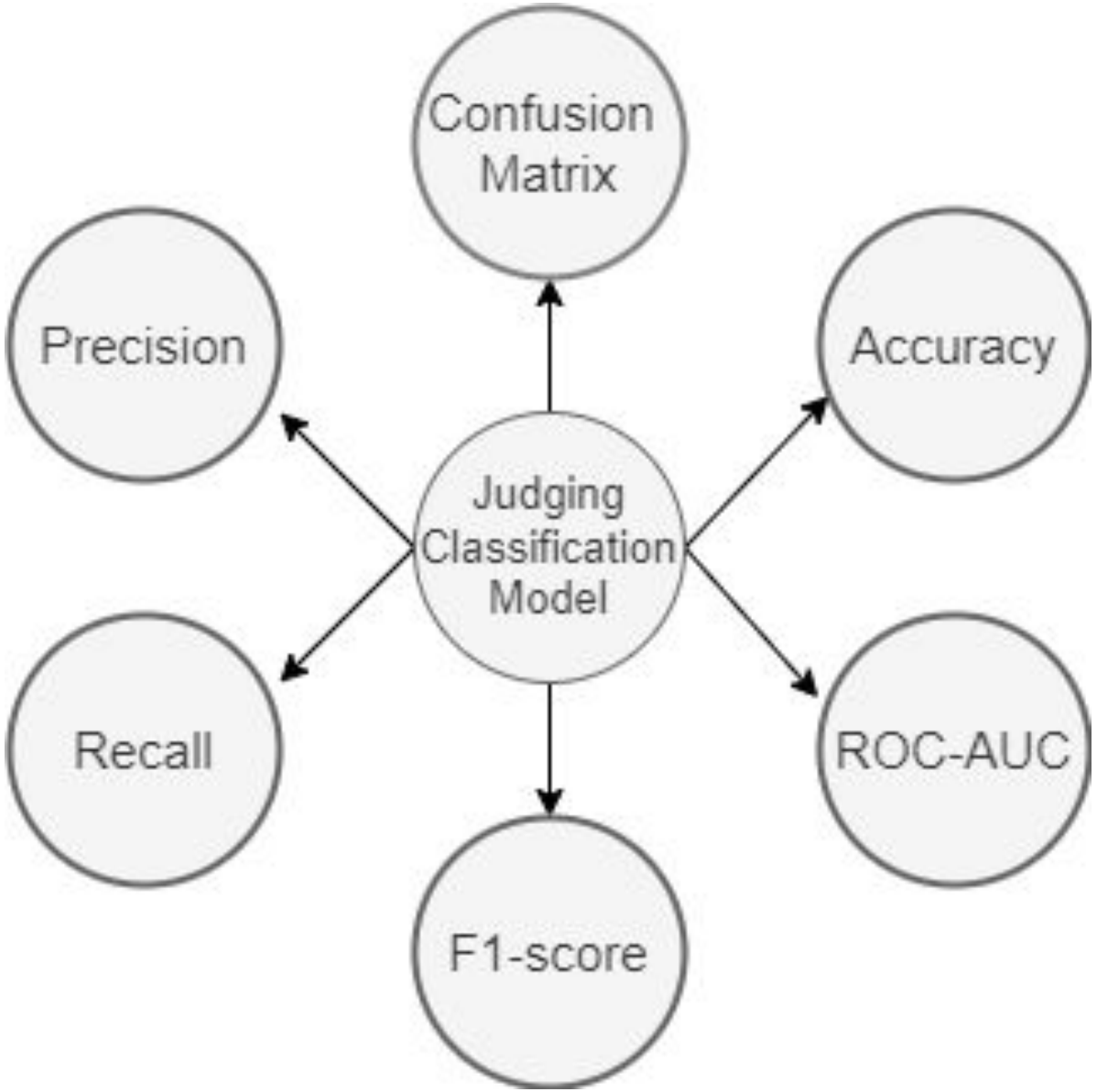
- Recap: Logistic Regression
- Evaluating Classification Models
  - Demo and Activity
- Interpreting Logistic Regression Models
  - **Activity**: Interpreting Coefficients and Log Odds
- Other Classification Algorithms
  - Naïve Bayes
  - Discriminant Analysis
- Multi-class Classification





Regression and Classification in Context of Statistical Learning


Prediction Evaluation
Statistical
Computational
P-values
Adj R-squared
F-stat
ANOVA
RMSE, MAE, R-squared
Predicted/actual fit
K-Fold Cross Validation
Confusion Matrix


Confusion Matrix
Precision
Accuracy
Judging Classification Model
Recall
ROC-AUC
F1-score

Confusion Matrix and Key Metrics

	Predicted		
Actual	Positive	Negative	Row Totals
Positive	TP	FN	TotActPos
Negative	FP	TN	TotActNeg
Col Totals	TotPredPos	TotPredNeg	Total

$$\text{Recall} = \frac{\text{TP}}{\text{TotActPos}}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TotAcNeg}}$$

$$\text{Error} = \frac{\text{\# Wrong}}{\text{Total}}$$

$$\text{Accuracy} = \frac{\text{\# Right}}{\text{Total}}$$

$$F = 2 * \frac{\text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TotPredPos}}$$

Metric	Definition	Use Case
Accuracy	The proportion of correctly classified instances (both true positive and true negative) over all instances.	Measures the overall performance of a classifier
Precision	The proportion of correctly classified positive instances over all instances that are classified as positive.	Measures the ability of the classifier to avoid false positives
Recall	The proportion of correctly classified positive instances over all actual positive instances.	Measures the ability of the classifier to identify all actual positive instances
F1-Score	The harmonic mean of precision and recall, providing a balanced measure of both precision and recall.	A good indicator of the performance of a classifier when the number of positive and negative instances is unbalanced

Types of errors

False positive rate: The fraction of negative examples that are classified as positive — 0.2% in example.

False negative rate: The fraction of positive examples that are classified as negative — 75.7% in example.

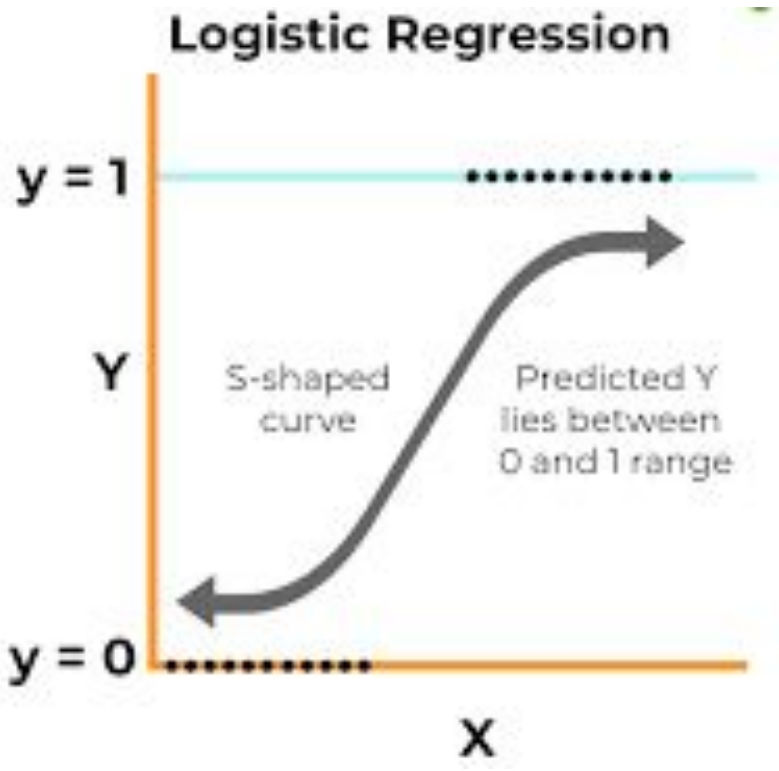
We produced this table by classifying to class `Yes` if

$$\widehat{\Pr}(\texttt{Default} = \texttt{Yes} | \texttt{Balance}, \texttt{Student}) \geq 0.5$$

We can change the two error rates by changing the threshold from 0.5 to some other value in $[0, 1]$:

$$\widehat{\Pr}(\texttt{Default} = \texttt{Yes} | \texttt{Balance}, \texttt{Student}) \geq \mathit{threshold},$$

and vary *threshold*.



a.k.a. **Log Odds**

or **Logit**

Intercept

$$\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X$$

Logistic Regression

- Logistic regression is a GLM used to model binary categorical outcome using numerical and categorical predictors
- The logistic regression is specified by
  - Bernouilli distribution
  - Linear function
  - Logit function (the linking function)

$$p_i = \frac{\exp(\beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i})}{1 + \exp(\beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i})}$$

$$Odds = \frac{\#\text{for}}{\#\text{against}} \qquad P = \frac{\#\text{for}}{\#\text{for} + \#\text{against}} = \frac{\#\text{for}}{\#\text{total}}$$

$$\#\text{total} = \#\text{for} + \#\text{against}$$

$$odds = \frac{P}{1-P}$$

a.k.a. **Log Odds**

or **Logit**

Intercept

$$\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X$$

$$\frac{P}{1-P} = e^{\beta_0 + \beta_1 X}$$

Intercept: the log of the odds of having the outcome, when X=0. The probability of having the outcome is $P = e^{\beta 0} / (1 + e^{\beta 0})$.

Coefficient: increasing the predictor by 1 unit multiplies the odds of having the outcome by e^{β}

The **log-odds** (or $\text{logit}(p)$) is the result of the linear part of the logistic regression model.

- A **positive** log-odds means the odds are greater than 1, so the probability p is greater than 0.5 (the outcome is **more likely** than not).
- A **negative** log-odds means the odds are less than 1, so the probability p is less than 0.5 (the outcome is **less likely** than not).
- A log-odds of **zero** means the odds are 1:1, so the probability p is exactly 0.5 (equally likely).

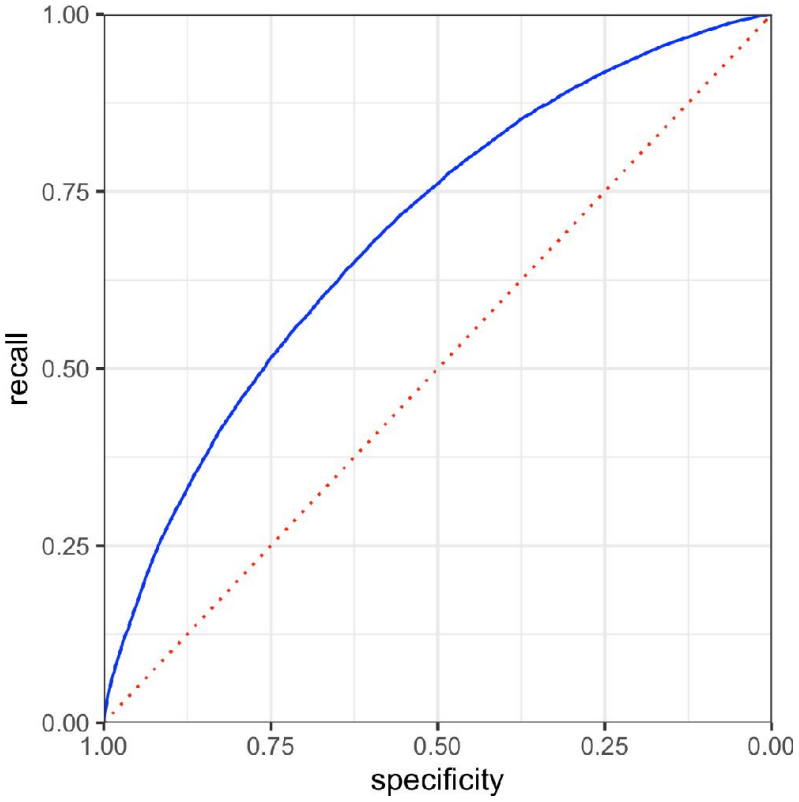
Example: Interpreting Logistic Regression Coefficients

Model:

$$\text{logit}(p) = 0.5 + 0.13 * \text{study_hours} + 0.97 * \text{female}$$

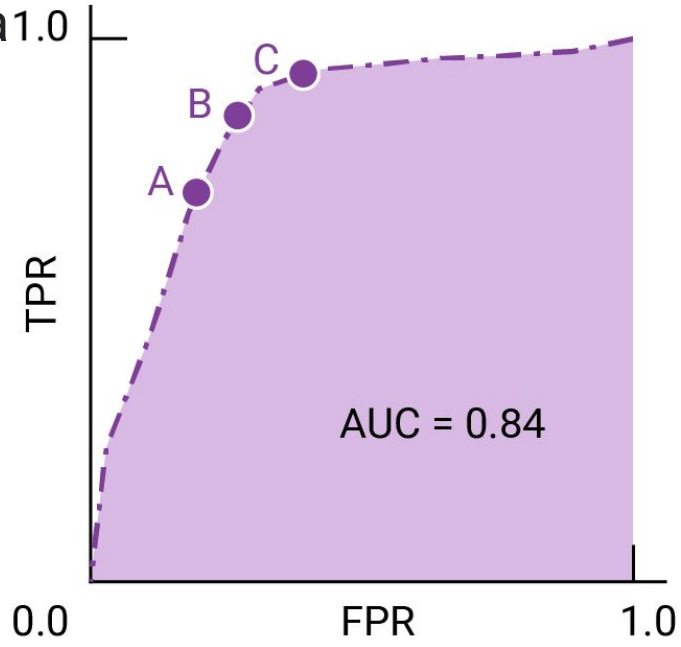
p = P{Y=1} is the probability of passing a math exam. Increasing the study hours by 1 unit (1 hour) will result in a 0.13 increase in *logit(p)* or *log(p/1-p)*. If *log(p/1–p)* increases by 0.13, that means that *p/(1 — p)* will increase by *exp(0.13) = 1.14*. This is a 14% increase in the odds of passing the exam (*assuming that the variable female remains fixed*). Assuming study hours is fixed, the impact of being female is given by *exp(0.97) = 2.64*. The odds for passing the exam are 164% higher for women.

Source: https://towardsdatascience.com/a-simple-interpretation-of-logistic-regression-coefficients-e3a40a62e8cf



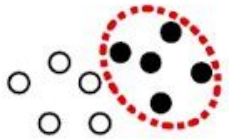
ROC Curve & AUC

- Captures the trade-off between **recall (sensitivity)** and **specificity**
- The model with greater area under the curve is generally the better one.
- Classification Threshold depends on which metric is most important for the use case

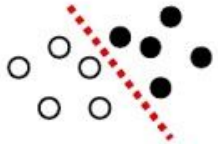


Generative vs. Discriminative

- Generative:
  - probabilistic "model" of each class
  - decision boundary:
    - where one model becomes more likely
  - natural use of unlabeled data



- Discriminative:
  - focus on the decision boundary
  - more powerful with lots of examples
  - not designed to use unlabeled data
  - only supervised tasks



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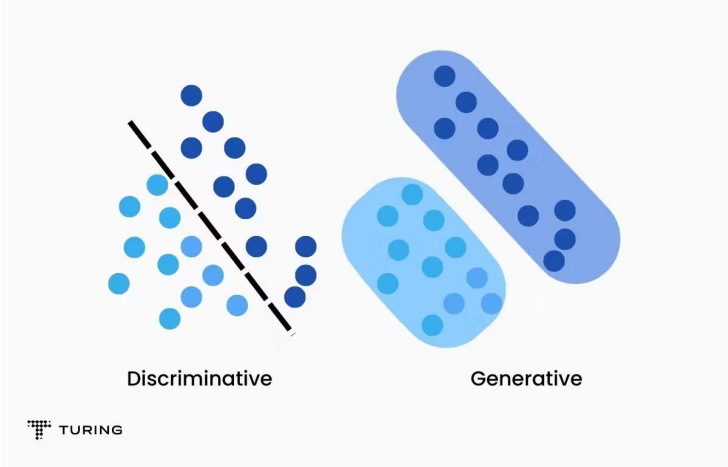
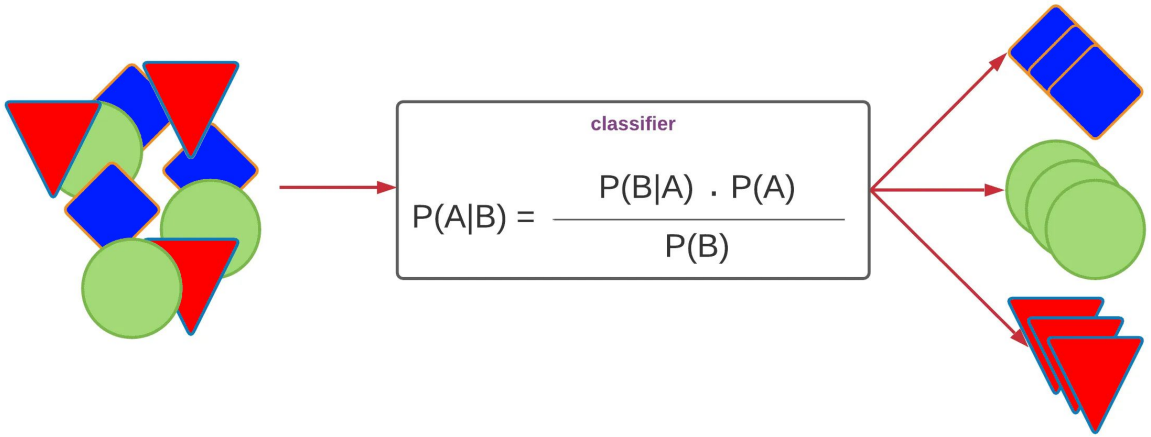


Comparison of methodologies

Model	Generative	Discriminative
Goal	Probability estimates	Classification rule
Performance measure	Likelihood	Misclassification rate
Mismatch problems	Outliers	Misclassifications

Naive Bayes Classifier



Key Terms for Naive Bayes

Conditional probability
The probability of observing some event (say, $X = i$) given some other event (say, $Y = i$), written as $P(X_i | Y_i)$.

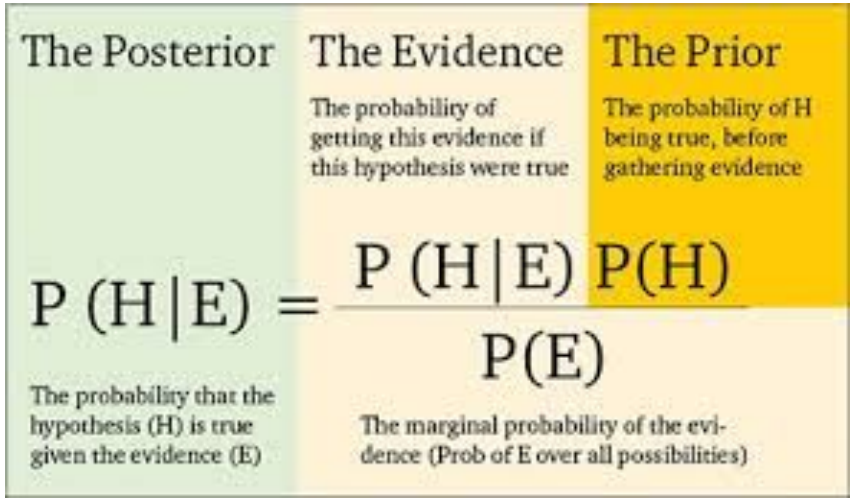
Posterior probability
The probability of an outcome after the predictor information has been incorporated (in contrast to the *prior probability* of outcomes, not taking predictor information into account).

Key Ideas

- Naive Bayes works with categorical (factor) predictors and outcomes.
- It asks, "Within each outcome category, which predictor categories are most probable?"
- That information is then inverted to estimate probabilities of outcome categories, given predictor values.

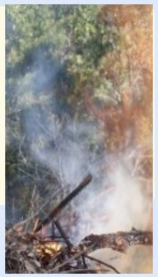
Naïve Bayes

Bayes' Theorem: How computers "learn" about people



Example:

- dangerous fires are rare (1%)
- but smoke is fairly common (10%) due to barbecues,
- and 90% of dangerous fires make smoke



Home Bonfire

We can then discover the **probability of dangerous Fire when there is Smoke**:

$$P(\text{Fire}|\text{Smoke}) = \frac{P(\text{Fire})\,P(\text{Smoke}|\text{Fire})}{P(\text{Smoke})} = \frac{1\% \times 90\%}{10\%} = 9\%$$

So it is still worth checking out any smoke to be sure.

Let us say P(Fire) means how often there is fire, and P(Smoke) means how often we see smoke, then:

P(Fire|Smoke) means how often there is fire when we can see smoke

P(Smoke|Fire) means how often we can see smoke when there is fire

So the formula kind of tells us "forwards" P(Fire|Smoke) when we know "backwards" P(Smoke|Fire)

Source: https://www.mathsisfun.com/data/bayes-theorem.html

Generative Models and Naïve Bayes

- Logistic regression models $\Pr(Y = k|X = x)$ directly, via the logistic function. Similarly the multinomial logistic regression uses the softmax function. These all model the *conditional distribution* of Y given X.
- By contrast *generative models* start with the conditional distribution of X given Y, and then use *Bayes formula* to turn things around:

$$\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}.$$

  - $f_k(x)$ is the density of X given $Y = k$;
  - $\pi_k = \Pr(Y = k)$ is the marginal probability that Y is in class k.

Naïve Bayes — Details

Why the independence assumption?

- Difficult to specify and model high-dimensional densities. Much easier to specify one-dimensional densities.
- Can handle *mixed* features:
  - If feature j is quantitative, can model as univariate Gaussian, for example: $X_j|Y = k \sim N(\mu_{jk}, \sigma^2_{jk})$. We estimate μ_{jk} and σ^2_{jk} from the data, and then plug into Gaussian density formula for $f_{jk}(x_j)$.
  - Alternatively, can use a *histogram* estimate of the density, and directly estimate $f_{jk}(x_j)$ by the proportion of observations in the bin into which x_j falls.
  - If feature j is qualitative, can simply model the proportion in each category. Example to follow.
- Somewhat unrealistic but extremely useful in many cases. Despite its simplicity, often shows good classification performance due to reduced variance.

Naïve Bayes vs. Logistic Regression (1)

NAÏVE BAYES (GENERATIVE)

- Assumes features are independent of each other
- Simple, computationally efficient
- Well-suited for categorical data; effective for discrete features and text classification
- Highly interpretable due to simplicity and assumption of feature independence

LOGISTIC REGRESSION (DISCRIMINATIVE)

- No assumption about independence
- More complex, allows modeling intricate relationships. Potential for overfitting with higher number of features
- Versatile, can handle both numerical and categorical features. Requires encoding for categorical variables with multiple levels
- Good interpretability through coefficients, indicating the strength and direction of feature relationships

Source: https://www.geeksforgeeks.org/naive-bayes-vs-logistic-regression-in-machine-learning/

Naïve Bayes vs. Logistic Regression (2)

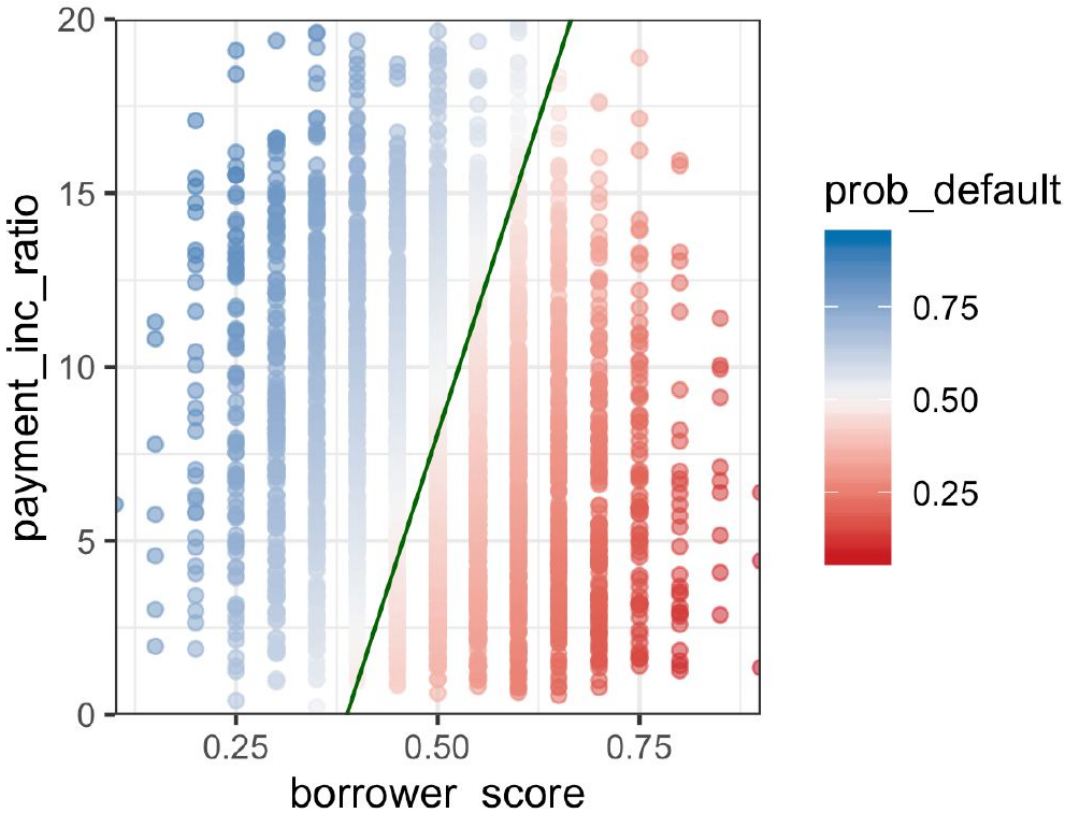
NB (GENERATIVE)

- Can be robust to irrelevant features due to the assumption of independence
- Performs well with small datasets. Effective with sparse data, e.g., in text classification
- Suitable for text classification, spam filtering, sentiment analysis

LOGISTIC REGRESSION (DISCRIMINATIVE)

- May be sensitive to irrelevant features. Regularization techniques can help mitigate this sensitivity.
- May require a larger dataset to avoid overfitting, especially with a high number of features. Can handle sparse data with careful regularization.
- Suitable for customer churn prediction, credit scoring

Discriminant Analysis



Key Terms for Discriminant Analysis

Covariance
A measure of the extent to which one variable varies in concert with another (i.e., similar magnitude and direction).

Discriminant function
The function that, when applied to the predictor variables, maximizes the separation of the classes.

Discriminant weights
The scores that result from the application of the discriminant function and are used to estimate probabilities of belonging to one class or another.

Key Ideas

- Discriminant analysis works with continuous or categorical predictors, as well as with categorical outcomes.
- Using the covariance matrix, it calculates a *linear discriminant function*, which is used to distinguish records belonging to one class from those belonging to another.
- This function is applied to the records to derive weights, or scores, for each record (one weight for each possible class), which determines its estimated class.

Discriminant Analysis

Here the approach is to model the distribution of X in each of the classes separately, and then use *Bayes theorem* to flip things around and obtain $\Pr(Y|X)$.

When we use normal (Gaussian) distributions for each class, this leads to linear or quadratic discriminant analysis.

However, this approach is quite general, and other distributions can be used as well. We will focus on normal distributions.

Why discriminant analysis?

- When the classes are well-separated, the parameter estimates for the logistic regression model are surprisingly unstable. Linear discriminant analysis does not suffer from this problem.
- If n is small and the distribution of the predictors X is approximately normal in each of the classes, the linear discriminant model is again more stable than the logistic regression model.
- Linear discriminant analysis is popular when we have more than two response classes, because it also provides low-dimensional views of the data.

Summary

- Logistic regression is very popular for classification, especially when there are two classes in the target variable.
- Naïve Bayes is useful for very large dimension data (many predictors).
- LDA is useful for when n is small, or the classes are well separated, and Gaussian assumptions are reasonable. Also when the target has multiple classes.